

Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj

2nd Year 1st Semester B. Sc. (Engg.) Second In-course Examination, 2023

Department of Computer Science and Engineering

CSE 2105: Computer Architecture (3 Credits)

Time: 1 Hour

Marks: 25

Answer all the questions

[The figures in the right margin indicate the marks. All portions of each question must be answered sequentially.]

1.	(a) Provide scenarios where overflow in addition and subtraction can occur. Explain with proper examples.	04
	(b) State how computer designers ignore the overflow in some cases, and recognise it in others.	04
2.	Draw the schematic diagram of the organisation of a simple computer with one CPU and two I/O devices.	03
3.	For the following C statement find the corresponding MIPS assembly code. Assume that variables i, j, k, l, m, n, and p are assigned to registers \$s1, \$s2, \$s3, \$s4, \$s5, \$s6, and \$s7, respectively. <pre>if (i==j) { if (k==l) { m=n+p; } else { m=n-p; } } else { m=p-n; }</pre>	06
4.	(a) State the advantages and disadvantages of RISC architecture.	04
	(b) "The CPUs in a Multiprocessor architecture are tightly coupled, while the CPUs in a Multicomputer architecture are loosely coupled"- justify the statement with proper explanation and suitable diagrams.	04